Named value indices in Clean

Named mathematical values can index types. The topology and endpoints below are explicit parameters; no ambient topology is selected implicitly.

Result 1 (subset element reflexivity). [Topology foundations, subset element reflexivity]
Let X be a type. Let S be a subset of X. Let x be an element of S. Then

$$x = x.$$

Result 2 (path reflexivity). [Topology foundations, path reflexivity]
Let X be a type. Let t be a topology on X. Let x and y be elements of X. Let p be a path from x to y in X with topology t. Then

$$p = p.$$

Result 3 (constant path exists). [Topology foundations, constant path exists]
Let X be a type. Let t be a topology on X. Then

$$(\text{for every } x : X,\ (\text{there exists } p : \mathrm{Path}_t(x, x),\ p = p))\,.$$

Result 4 (named subset expression). [Topology foundations, named subset expression]
Let X and Y be types. Let f be a function from X to Y. Let S be a subset of Y. Let $T = f^{-1}(S)$. Let x be an element of T. Then

$$x = x.$$